

CRC Safety Rules

Highlights taken from the 2001 R&D Chemical Hygiene Plan (O:\COU_ReseDevel_Safety\HYGIENE)

AWARENESS IS THE SINGLE MOST FUNDAMENTAL RULE OF CHEMICAL SAFETY

Emergency Procedures

Large Chemical Spill or Toxic Gas Release

Evacuate lab, Dial "O", describe emergency and location to operator, evacuate building if necessary. Operator activates emergency response team.

Chemical exposure:

Get to safety shower/eyewash ASAP, flush affected body with water for at least 15 minutes. If alone, activate emergency button, otherwise co-worker dials "O" and describes emergency and location. Emergency response team will be activated and respond.

Facility Alarms:

Continuous: Fire or Chemical Emergency

Intermittent: Security

Lab Alarm: Flashing light and audible alarm above lab entrance

Chemical Handling and Storage

- All personnel handling chemicals must read MSDS and be fully aware of all hazards.
- Properly label all chemical containers in lab.
- MSDS in lab required for all chemicals used or stored in lab.
- Label chemicals and gas cylinders in storage using group color codes.
- Flammable solvents are to be stored in the chemical storeroom.
- Individual solvents in use in the lab must not exceed 1-liter in volume.
- Individual chemicals in a container and less than 15 lbs can be transported using rubber "bucket-type" carriers.
- Specific rules concerning transportation of multiple chemicals and larger amounts are given in the CHP.

Chemical Waste and Disposal

- Waste generation is expensive, dangerous and must be minimized.
- Restrict purchase, storage and use of all chemicals to the absolute minimum required.
- Do not dispose of hazardous chemicals in the sink, drain, wastebasket, or hood.
- Organic solvents, halogenated solvents and waste oil can be disposed of in the chemical storeroom in the appropriate 55-gallon drums.
- If a chemical can be made non-hazardous (such as neutralizing acids and bases), follow the recommended practice and dispose of properly in the sink, drain or wastebasket. See "Prudent Practices for Disposal of Chemicals in Laboratories"
- Chemical waste that cannot be made non-hazardous must be disposed of using one of the Air Liquide- approved waste vendors.
- Contact chemical storage coordinator for more specific information.

Gas Cylinder Handling & Storage

- Gas cylinders must be transported using 1-cylinder, 4-wheel carts.
- The cylinder cap must be on the gas cylinder during transportation.
- Inert gases (N₂, Ar, He, compressed air, CO₂ are maintained in a general cylinder inventory inside the facility.
- Toxic, flammable, oxidizing gases and gas mixtures are stored outside in the cylinder storage area and labeled with the group color code.
- Follow common procedures when connecting regulators and gas delivery systems to high-pressure cylinders.

General Safety Rules

- No eating or drinking in labs.
- Safety glasses and safety shoes required in laboratories.
- Working alone in laboratories after hours or on weekends is prohibited.
- Practice good laboratory housekeeping.
- MSDS sheets required for all chemicals
 - <http://www.siri.org/msds/index.html> (one web source)
 - <http://www.airliquide.com> in Quick Links → MSDS
- Work permits required for hazardous lab experiments and procedures.
- Be sure you have been properly trained before attempting lab procedures.
- Customer safety plan required for offsite work.

Personal Hygiene Practices

- Wash promptly if skin contact made with chemicals.
- Wash hands with soap after leaving the laboratory.
- Remove lab coat after leaving laboratory; launder work clothes often.
- Avoid inhalation of chemicals; don't "sniff-test" chemicals.
- Don't mouth pipette anything.

Hazardous Condition Safeguards

- Wear clothing that minimizes exposed skin.
- Use appropriate personal protective equipment for hazardous conditions:
 - Lab coats, gloves, face shields, ear protection, splash aprons.
- Respirators are generally not used for normal procedures, periodic use is OK but must be fit-tested annually.
- Use fume hoods for open air handling of chemicals with threshold limit values (TLV) less than 1000 ppm.
- Make sure fume hood is working properly before use and is rated correctly for the specific chemicals (flammable, toxic).
- Don't use odor as primary method of vapor detection.
- Use safety monitors if needed; Periodically check that they are working correctly.

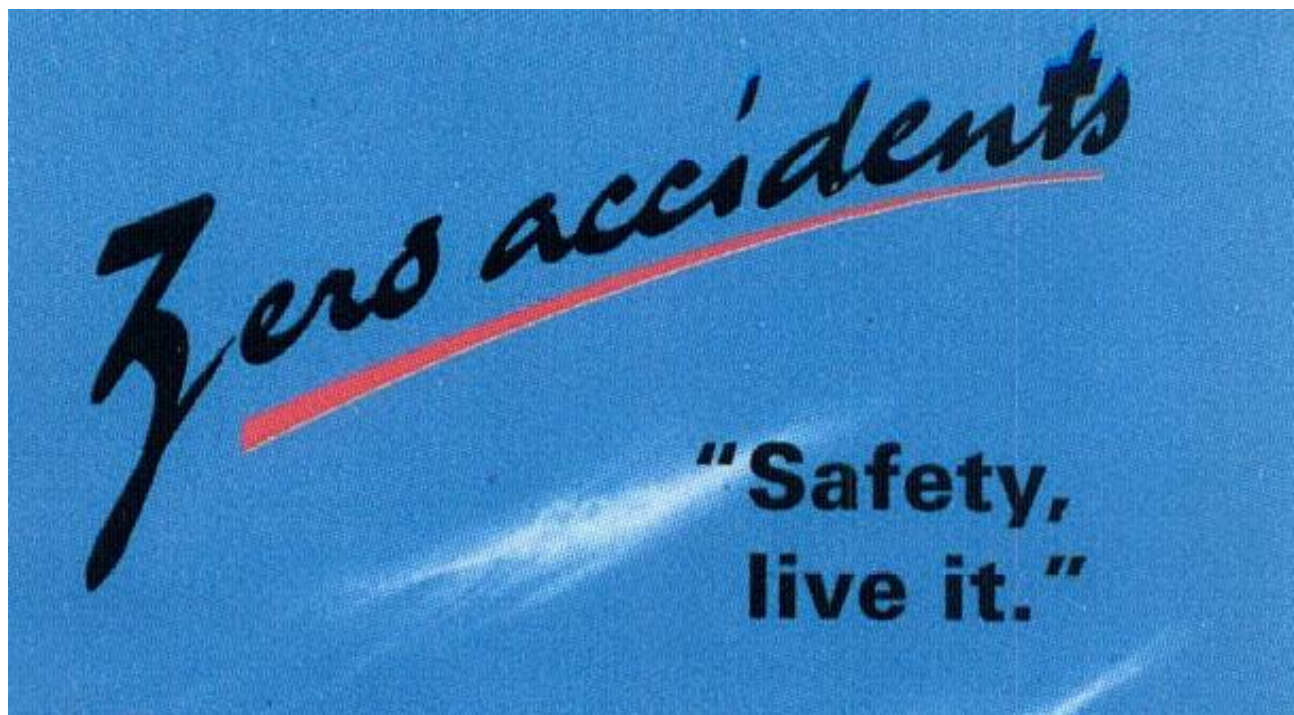
Miscellaneous Rules & Practices

- Components to be used in oxygen service must be "Cleaned for Oxygen Service" following standard practices as outlined by Air Liquide America policy. See the Chemical Hygiene Officer for more specific information.
- Special precautions for storage, use, and disposal of carcinogens are required. Please see the carcinogen chemical coordinator for more information.

2001 FACILITY SAFETY TEAM

- Facility Manager: Ron Nowak (7797)
- CRC Chemical Hygiene Officer: Alan Zdunek (7864)
- ALTEC Safety Officer: Laura Dovalovsky (7912)
- MW/NE Region Safety Officer: Don Groves (7736)
- Cylinder Inventory: Rick Udischas (7763) & Roy Chamcharas (7743)
- Chemical Storage & Disposal: Alex Barajas (7747) & Tay Sayasane (7756)
- Work Permit Coordinator: Mindi Xu (7884)
- Carcinogen Chemical Coordinator: Michelle Kelly (7714)
- Fume Hood Inspector: Sejal Thakkar (7730)

**REMEMBER: NO AMOUNT OF INSURANCE CAN CURE BLINDNESS OR RESURRECT THE DEAD!
BE RESPONSIBLE FOR YOUR OWN WELL-BEING. IF YOU'RE NOT SURE ABOUT SOMETHING,
ASK YOUR SUPERVISOR OR OTHER EXPERIENCED PERSON TO HELP YOU!**



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